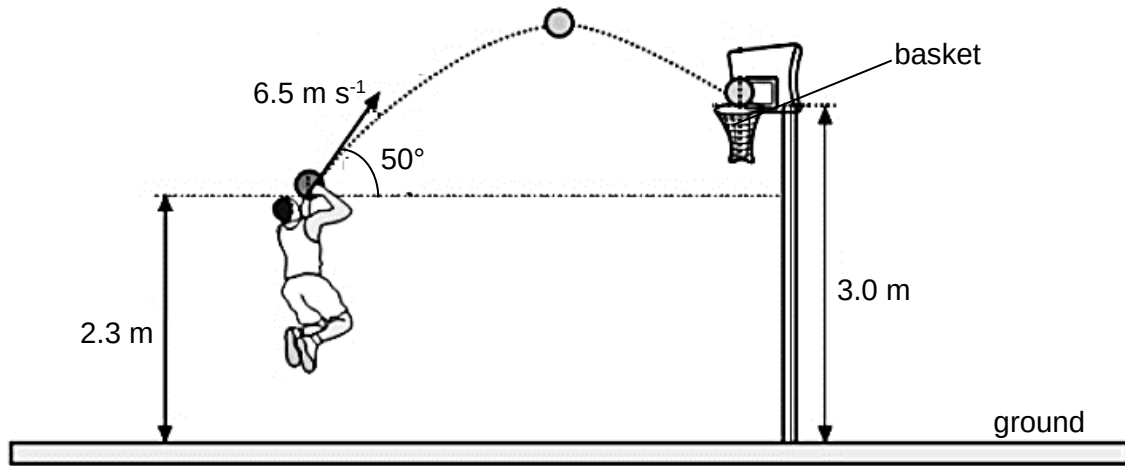


2

A basketball player throws a ball with an initial velocity of 6.5 m s^{-1} at an angle of 50° to the horizontal. The ball is 2.3 m above the ground when released and passes through the basket on its way down.



	What is the time taken for the ball to reach the top of the basket which is 3.0 m above the ground?							
	A	0.17 s	B	0.36 s	C	0.54 s	D	0.85 s
L2	Answer: D Take upwards as positive direction. $s_y = u_y t + \frac{1}{2} a_y t^2$ $(3.0 - 2.3) = (6.5 \sin 50^\circ) t + \frac{1}{2} (-9.81) t^2$ $0.7 = 4.97929t - 4.905t^2$ $4.905t^2 - 4.97929t + 0.7 = 0$ $t = 0.16858s(\text{reject}) \text{ or } 0.84657s$ $= 0.85s \text{ (2 s.f.)}$ Accept the answer where the ball reaches displacement of 0.7 m the 2nd time.							

2	An object of mass 20 kg moves along a straight line on a smooth horizontal surface. A force of 5 N acts on the object in the direction of motion. The object starts from rest and moves a distance of 10 m. Find the speed of the object at the end of this distance.					
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